

Finally, the relationships between (A_x, A_y, A_z) and (A_ρ, A_ϕ, A_z) are obtained by simply substituting eq. (2.9) into eq. (2.2) and collecting terms. Thus

$$\mathbf{A} = (A_x \cos \phi + A_y \sin \phi)\mathbf{a}_\rho + (-A_x \sin \phi + A_y \cos \phi)\mathbf{a}_\phi + A_z\mathbf{a}_z \quad (2.11)$$

or

$$\begin{aligned} A_\rho &= A_x \cos \phi + A_y \sin \phi \\ A_\phi &= -A_x \sin \phi + A_y \cos \phi \\ A_z &= A_z \end{aligned} \quad (2.12)$$

In matrix form, we have the transformation of vector $\mathbf{A}$ from (A_x, A_y, A_z) to (A_ρ, A_ϕ, A_z) as

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} \quad (2.13)$$

The inverse of the transformation $(A_\rho, A_\phi, A_z) \rightarrow (A_x, A_y, A_z)$ is obtained as

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} \quad (2.14)$$

or directly from eqs. (2.4) and (2.10). Thus

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} \quad (2.15)$$

An alternative way of obtaining eq. (2.14) or (2.15) is using the dot product. For example:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \mathbf{a}_x \cdot \mathbf{a}_\rho & \mathbf{a}_x \cdot \mathbf{a}_\phi & \mathbf{a}_x \cdot \mathbf{a}_z \\ \mathbf{a}_y \cdot \mathbf{a}_\rho & \mathbf{a}_y \cdot \mathbf{a}_\phi & \mathbf{a}_y \cdot \mathbf{a}_z \\ \mathbf{a}_z \cdot \mathbf{a}_\rho & \mathbf{a}_z \cdot \mathbf{a}_\phi & \mathbf{a}_z \cdot \mathbf{a}_z \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} \quad (2.16)$$

The derivation of this is left as an exercise.

2.4 SPHERICAL COORDINATES (r, θ, ϕ)

The spherical coordinate system is most appropriate when dealing with problems having a degree of spherical symmetry. A point P can be represented as (r, θ, ϕ) and is illustrated in Figure 2.4. From Figure 2.4, we notice that r is defined as the distance from the origin to

point P or the radius of a sphere centered at the origin and passing through P ; θ (called the *colatitude*) is the angle between the z -axis and the position vector of P ; and ϕ is measured from the x -axis (the same azimuthal angle in cylindrical coordinates). According to these definitions, the ranges of the variables are

$$\begin{aligned} 0 &\leq r < \infty \\ 0 &\leq \theta \leq \pi \\ 0 &\leq \phi < 2\pi \end{aligned} \quad (2.17)$$

A vector $\mathbf{A}$ in spherical coordinates may be written as

$$(A_r, A_\theta, A_\phi) \quad \text{or} \quad A_r \mathbf{a}_r + A_\theta \mathbf{a}_\theta + A_\phi \mathbf{a}_\phi \quad (2.18)$$

where $\mathbf{a}_r$, $\mathbf{a}_\theta$, and $\mathbf{a}_\phi$ are unit vectors along the r -, θ -, and ϕ -directions. The magnitude of $\mathbf{A}$ is

$$|\mathbf{A}| = (A_r^2 + A_\theta^2 + A_\phi^2)^{1/2} \quad (2.19)$$

The unit vectors $\mathbf{a}_r$, $\mathbf{a}_\theta$, and $\mathbf{a}_\phi$ are mutually orthogonal; $\mathbf{a}_r$ being directed along the radius or in the direction of increasing r , $\mathbf{a}_\theta$ in the direction of increasing θ , and $\mathbf{a}_\phi$ in the direction of increasing ϕ . Thus,

$$\begin{aligned} \mathbf{a}_r \cdot \mathbf{a}_r &= \mathbf{a}_\theta \cdot \mathbf{a}_\theta = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1 \\ \mathbf{a}_r \cdot \mathbf{a}_\theta &= \mathbf{a}_\theta \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_r = 0 \\ \mathbf{a}_r \times \mathbf{a}_\theta &= \mathbf{a}_\phi \\ \mathbf{a}_\theta \times \mathbf{a}_\phi &= \mathbf{a}_r \\ \mathbf{a}_\phi \times \mathbf{a}_r &= \mathbf{a}_\theta \end{aligned} \quad (2.20)$$

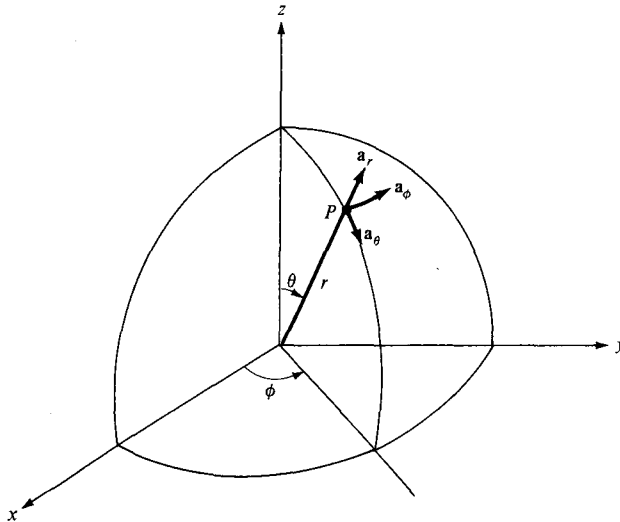


Figure 2.4 Point P and unit vectors in spherical coordinates.

The space variables (x, y, z) in Cartesian coordinates can be related to variables (r, θ, ϕ) of a spherical coordinate system. From Figure 2.5 it is easy to notice that

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z}, \quad \phi = \tan^{-1} \frac{y}{x} \quad (2.21)$$

or

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta \quad (2.22)$$

In eq. (2.21), we have $(x, y, z) \rightarrow (r, \theta, \phi)$ point transformation and in eq. (2.22), it is $(r, \theta, \phi) \rightarrow (x, y, z)$ point transformation.

The unit vectors $\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z$ and $\mathbf{a}_r, \mathbf{a}_\theta, \mathbf{a}_\phi$ are related as follows:

$$\begin{aligned} \mathbf{a}_x &= \sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi \\ \mathbf{a}_y &= \sin \theta \sin \phi \mathbf{a}_r + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi \\ \mathbf{a}_z &= \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_\theta \end{aligned} \quad (2.23)$$

or

$$\begin{aligned} \mathbf{a}_r &= \sin \theta \cos \phi \mathbf{a}_x + \sin \theta \sin \phi \mathbf{a}_y + \cos \theta \mathbf{a}_z \\ \mathbf{a}_\theta &= \cos \theta \cos \phi \mathbf{a}_x + \cos \theta \sin \phi \mathbf{a}_y - \sin \theta \mathbf{a}_z \\ \mathbf{a}_\phi &= -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y \end{aligned} \quad (2.24)$$

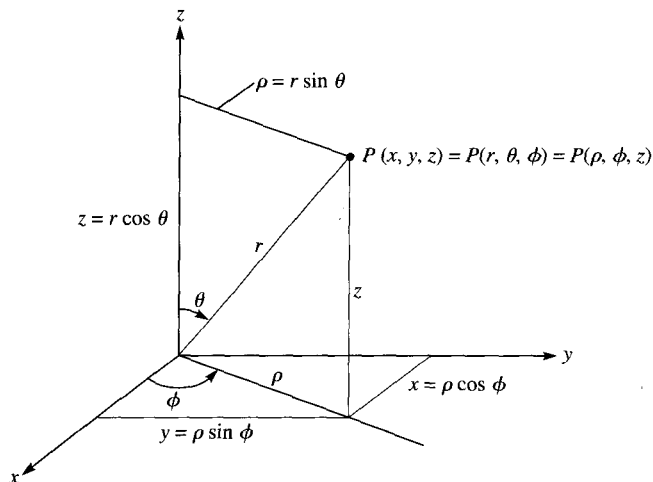


Figure 2.5 Relationships between space variables (x, y, z) , (r, θ, ϕ) , and (ρ, ϕ, z) .

The components of vector $\mathbf{A} = (A_x, A_y, A_z)$ and $\mathbf{A} = (A_r, A_\theta, A_\phi)$ are related by substituting eq. (2.23) into eq. (2.2) and collecting terms. Thus,

$$\mathbf{A} = (A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta) \mathbf{a}_r + (A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta) \mathbf{a}_\theta + (-A_x \sin \phi + A_y \cos \phi) \mathbf{a}_\phi \quad (2.25)$$

and from this, we obtain

$$\begin{aligned} A_r &= A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta \\ A_\theta &= A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta \\ A_\phi &= -A_x \sin \phi + A_y \cos \phi \end{aligned} \quad (2.26)$$

In matrix form, the $(A_x, A_y, A_z) \rightarrow (A_r, A_\theta, A_\phi)$ vector transformation is performed according to

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ -\cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} \quad (2.27)$$

The inverse transformation $(A_r, A_\theta, A_\phi) \rightarrow (A_x, A_y, A_z)$ is similarly obtained, or we obtain it from eq. (2.23). Thus,

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} \quad (2.28)$$

Alternatively, we may obtain eqs. (2.27) and (2.28) using the dot product. For example,

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \mathbf{a}_r \cdot \mathbf{a}_x & \mathbf{a}_r \cdot \mathbf{a}_y & \mathbf{a}_r \cdot \mathbf{a}_z \\ \mathbf{a}_\theta \cdot \mathbf{a}_x & \mathbf{a}_\theta \cdot \mathbf{a}_y & \mathbf{a}_\theta \cdot \mathbf{a}_z \\ \mathbf{a}_\phi \cdot \mathbf{a}_x & \mathbf{a}_\phi \cdot \mathbf{a}_y & \mathbf{a}_\phi \cdot \mathbf{a}_z \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} \quad (2.29)$$

For the sake of completeness, it may be instructive to obtain the point or vector transformation relationships between cylindrical and spherical coordinates using Figures 2.5 and 2.6 (where ϕ is held constant since it is common to both systems). This will be left as an exercise (see Problem 2.9). Note that in point or vector transformation the point or vector has not changed; it is only expressed differently. Thus, for example, the magnitude of a vector will remain the same after the transformation and this may serve as a way of checking the result of the transformation.

The distance between two points is usually necessary in EM theory. The distance d between two points with position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ is generally given by

$$d = |\mathbf{r}_2 - \mathbf{r}_1| \quad (2.30)$$

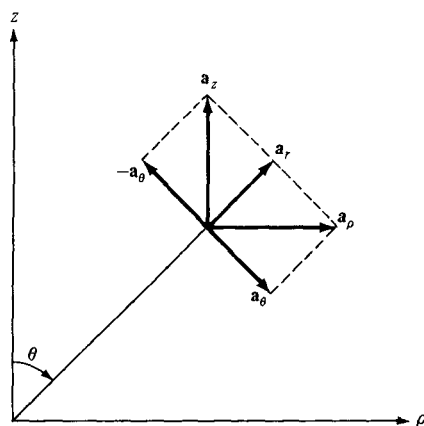


Figure 2.6 Unit vector transformations for cylindrical and spherical coordinates.

or

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 \text{ (Cartesian)} \quad (2.31)$$

$$d^2 = \rho_2^2 + \rho_1^2 - 2\rho_1\rho_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2 \text{ (cylindrical)} \quad (2.32)$$

$$d^2 = r_2^2 + r_1^2 - 2r_1r_2 \cos \theta_2 \cos \theta_1 - 2r_1r_2 \sin \theta_2 \sin \theta_1 \cos(\phi_2 - \phi_1) \text{ (spherical)} \quad (2.33)$$

EXAMPLE 2.1

Given point $P(-2, 6, 3)$ and vector $\mathbf{A} = y\mathbf{a}_x + (x + z)\mathbf{a}_y$, express P and $\mathbf{A}$ in cylindrical and spherical coordinates. Evaluate $\mathbf{A}$ at P in the Cartesian, cylindrical, and spherical systems.

Solution:

At point P : $x = -2, y = 6, z = 3$. Hence,

$$\rho = \sqrt{x^2 + y^2} = \sqrt{4 + 36} = 6.32$$

$$\phi = \tan^{-1} \frac{y}{x} = \tan^{-1} \frac{6}{-2} = 108.43^\circ$$

$$z = 3$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{4 + 36 + 9} = 7$$

$$\theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z} = \tan^{-1} \frac{\sqrt{40}}{3} = 64.62^\circ$$

Thus,

$$P(-2, 6, 3) = P(6.32, 108.43^\circ, 3) = P(7, 64.62^\circ, 108.43^\circ)$$

In the Cartesian system, $\mathbf{A}$ at P is

$$\mathbf{A} = 6\mathbf{a}_x + \mathbf{a}_y$$

For vector $\mathbf{A}$, $A_x = y$, $A_y = x + z$, $A_z = 0$. Hence, in the cylindrical system

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y \\ x + z \\ 0 \end{bmatrix}$$

or

$$A_\rho = y \cos \phi + (x + z) \sin \phi$$

$$A_\phi = -y \sin \phi + (x + z) \cos \phi$$

$$A_z = 0$$

But $x = \rho \cos \phi$, $y = \rho \sin \phi$, and substituting these yields

$$\begin{aligned} \mathbf{A} = (A_\rho, A_\phi, A_z) &= [\rho \cos \phi \sin \phi + (\rho \cos \phi + z) \sin \phi] \mathbf{a}_\rho \\ &\quad + [-\rho \sin^2 \phi + (\rho \cos \phi + z) \cos \phi] \mathbf{a}_\phi \end{aligned}$$

At P

$$\rho = \sqrt{40}, \quad \tan \phi = \frac{6}{-2}$$

Hence,

$$\cos \phi = \frac{-2}{\sqrt{40}}, \quad \sin \phi = \frac{6}{\sqrt{40}}$$

$$\begin{aligned} \mathbf{A} &= \left[\sqrt{40} \cdot \frac{-2}{\sqrt{40}} \cdot \frac{6}{\sqrt{40}} + \left(\sqrt{40} \cdot \frac{-2}{\sqrt{40}} + 3 \right) \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_\rho \\ &\quad + \left[-\sqrt{40} \cdot \frac{36}{40} + \left(\sqrt{40} \cdot \frac{-2}{\sqrt{40}} + 3 \right) \cdot \frac{-2}{\sqrt{40}} \right] \mathbf{a}_\phi \\ &= \frac{-6}{\sqrt{40}} \mathbf{a}_\rho - \frac{38}{\sqrt{40}} \mathbf{a}_\phi = -0.9487 \mathbf{a}_\rho - 6.008 \mathbf{a}_\phi \end{aligned}$$

Similarly, in the spherical system

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} y \\ x + z \\ 0 \end{bmatrix}$$

or

$$A_r = y \sin \theta \cos \phi + (x + z) \sin \theta \sin \phi$$

$$A_\theta = y \cos \theta \cos \phi + (x + z) \cos \theta \sin \phi$$

$$A_\phi = -y \sin \phi + (x + z) \cos \phi$$

But $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, and $z = r \cos \theta$. Substituting these yields

$$\begin{aligned}\mathbf{A} &= (A_r, A_\theta, A_\phi) \\ &= r[\sin^2 \theta \cos \phi \sin \phi + (\sin \theta \cos \phi + \cos \theta) \sin \theta \sin \phi] \mathbf{a}_r \\ &\quad + r[\sin \theta \cos \theta \sin \phi \cos \phi + (\sin \theta \cos \phi + \cos \theta) \cos \theta \sin \phi] \mathbf{a}_\theta \\ &\quad + r[-\sin \theta \sin^2 \phi + (\sin \theta \cos \phi + \cos \theta) \cos \phi] \mathbf{a}_\phi\end{aligned}$$

At P

$$r = 7, \quad \tan \phi = \frac{6}{-2}, \quad \tan \theta = \frac{\sqrt{40}}{3}$$

Hence,

$$\begin{aligned}\cos \phi &= \frac{-2}{\sqrt{40}}, \quad \sin \phi = \frac{6}{\sqrt{40}}, \quad \cos \theta = \frac{3}{7}, \quad \sin \theta = \frac{\sqrt{40}}{7} \\ \mathbf{A} &= 7 \cdot \left[\frac{40}{49} \cdot \frac{-2}{\sqrt{40}} \cdot \frac{6}{\sqrt{40}} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{\sqrt{40}}{7} \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_r \\ &\quad + 7 \cdot \left[\frac{\sqrt{40}}{7} \cdot \frac{3}{7} \cdot \frac{6}{\sqrt{40}} \cdot \frac{-2}{\sqrt{40}} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{3}{7} \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_\theta \\ &\quad + 7 \cdot \left[\frac{-\sqrt{40}}{7} \cdot \frac{36}{40} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{-2}{\sqrt{40}} \right] \mathbf{a}_\phi \\ &= \frac{-6}{7} \mathbf{a}_r - \frac{18}{7\sqrt{40}} \mathbf{a}_\theta - \frac{38}{\sqrt{40}} \mathbf{a}_\phi \\ &= -0.8571 \mathbf{a}_r - 0.4066 \mathbf{a}_\theta - 6.008 \mathbf{a}_\phi\end{aligned}$$

Note that $|\mathbf{A}|$ is the same in the three systems; that is,

$$|\mathbf{A}(x, y, z)| = |\mathbf{A}(\rho, \phi, z)| = |\mathbf{A}(r, \theta, \phi)| = 6.083$$

PRACTICE EXERCISE 2.1

(a) Convert points $P(1, 3, 5)$, $T(0, -4, 3)$, and $S(-3, -4, -10)$ from Cartesian to cylindrical and spherical coordinates.

(b) Transform vector

$$\mathbf{Q} = \frac{\sqrt{x^2 + y^2} \mathbf{a}_x}{\sqrt{x^2 + y^2 + z^2}} - \frac{yz \mathbf{a}_z}{\sqrt{x^2 + y^2 + z^2}}$$

to cylindrical and spherical coordinates.

(c) Evaluate $\mathbf{Q}$ at T in the three coordinate systems.

Answer: (a) $P(3.162, 71.56^\circ, 5)$, $P(5.916, 32.31^\circ, 71.56^\circ)$, $T(4, 270^\circ, 3)$,
 $T(5, 53.13^\circ, 270^\circ)$, $S(5, 233.1^\circ, -10)$, $S(11.18, 153.43^\circ, 233.1^\circ)$

$$(b) \frac{\rho}{\sqrt{\rho^2 + z^2}} (\cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi - z \sin \phi \mathbf{a}_z), \sin \theta (\sin \theta \cos \phi - r \cos^2 \theta \sin \phi) \mathbf{a}_r + \sin \theta \cos \theta (\cos \phi + r \sin \theta \sin \phi) \mathbf{a}_\theta - \sin \theta \sin \phi \mathbf{a}_\phi$$

$$(c) 0.8\mathbf{a}_x + 2.4\mathbf{a}_z, 0.8\mathbf{a}_x + 2.4\mathbf{a}_z, 1.44\mathbf{a}_r - 1.92\mathbf{a}_\theta + 0.8\mathbf{a}_\phi.$$

EXAMPLE 2.2

Express vector

$$\mathbf{B} = \frac{10}{r} \mathbf{a}_r + r \cos \theta \mathbf{a}_\theta + \mathbf{a}_\phi$$

in Cartesian and cylindrical coordinates. Find $\mathbf{B}(-3, 4, 0)$ and $\mathbf{B}(5, \pi/2, -2)$.

Solution:

Using eq. (2.28):

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} \frac{10}{r} \\ r \cos \theta \\ 1 \end{bmatrix}$$

or

$$B_x = \frac{10}{r} \sin \theta \cos \phi + r \cos^2 \theta \cos \phi - \sin \phi$$

$$B_y = \frac{10}{r} \sin \theta \sin \phi + r \cos^2 \theta \sin \phi + \cos \phi$$

$$B_z = \frac{10}{r} \cos \theta - r \cos \theta \sin \theta$$

But $r = \sqrt{x^2 + y^2 + z^2}$, $\theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z}$, and $\phi = \tan^{-1} \frac{y}{x}$

Hence,

$$\sin \theta = \frac{\rho}{r} = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}, \quad \cos \theta = \frac{z}{r} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\sin \phi = \frac{y}{\rho} = \frac{y}{\sqrt{x^2 + y^2}}, \quad \cos \phi = \frac{x}{\rho} = \frac{x}{\sqrt{x^2 + y^2}}$$

Substituting all these gives

$$B_x = \frac{10\sqrt{x^2 + y^2}}{(x^2 + y^2 + z^2)} \cdot \frac{x}{\sqrt{x^2 + y^2}} + \frac{\sqrt{x^2 + y^2 + z^2}}{(x^2 + y^2 + z^2)} \cdot \frac{z^2 x}{\sqrt{x^2 + y^2}} - \frac{y}{\sqrt{x^2 + y^2}}$$

$$= \frac{10x}{x^2 + y^2 + z^2} + \frac{xz^2}{\sqrt{(x^2 + y^2)(x^2 + y^2 + z^2)}} - \frac{y}{\sqrt{(x^2 + y^2)}}$$

$$B_y = \frac{10\sqrt{x^2 + y^2}}{(x^2 + y^2 + z^2)} \cdot \frac{y}{\sqrt{x^2 + y^2}} + \frac{\sqrt{x^2 + y^2 + z^2}}{(x^2 + y^2 + z^2)} \cdot \frac{z^2 y}{\sqrt{x^2 + y^2}} - \frac{y}{\sqrt{x^2 + y^2}}$$

$$= \frac{10y}{x^2 + y^2 + z^2} + \frac{yz^2}{\sqrt{(x^2 + y^2)(x^2 + y^2 + z^2)}} + \frac{x}{\sqrt{x^2 + y^2}}$$

$$B_z = \frac{10z}{x^2 + y^2 + z^2} - \frac{z\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}$$

$$\mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$$

where B_x , B_y , and B_z are as given above.

At $(-3, 4, 0)$, $x = -3$, $y = 4$, and $z = 0$, so

$$B_x = -\frac{30}{25} + 0 - \frac{4}{5} = -2$$

$$B_y = \frac{40}{25} + 0 - \frac{3}{5} = 1$$

$$B_z = 0 - 0 = 0$$

Thus,

$$\mathbf{B} = -2\mathbf{a}_x + \mathbf{a}_y$$

For spherical to cylindrical vector transformation (see Problem 2.9),

$$\begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} \frac{10}{r} \\ r \cos \theta \\ 1 \end{bmatrix}$$

or

$$B_\rho = \frac{10}{r} \sin \theta + r \cos^2 \theta$$

$$B_\phi = 1$$

$$B_z = \frac{10}{r} \cos \theta - r \sin \theta \cos \theta$$

But $r = \sqrt{\rho^2 + z^2}$ and $\theta = \tan^{-1} \frac{\rho}{z}$

Thus,

$$\sin \theta = \frac{\rho}{\sqrt{\rho^2 + z^2}}, \quad \cos \theta = \frac{z}{\sqrt{\rho^2 + z^2}}$$

$$B_\rho = \frac{10\rho}{\rho^2 + z^2} + \sqrt{\rho^2 + z^2} \cdot \frac{z^2}{\rho^2 + z^2}$$

$$B_z = \frac{10z}{\rho^2 + z^2} - \sqrt{\rho^2 + z^2} \cdot \frac{\rho z}{\rho^2 + z^2}$$

Hence,

$$\mathbf{B} = \left(\frac{10\rho}{\rho^2 + z^2} + \frac{z^2}{\sqrt{\rho^2 + z^2}} \right) \mathbf{a}_\rho + \mathbf{a}_\phi + \left(\frac{10z}{\rho^2 + z^2} - \frac{\rho z}{\sqrt{\rho^2 + z^2}} \right) \mathbf{a}_z$$

At $(5, \pi/2, -2)$, $\rho = 5$, $\phi = \pi/2$, and $z = -2$, so

$$\begin{aligned} \mathbf{B} &= \left(\frac{50}{29} + \frac{4}{\sqrt{29}} \right) \mathbf{a}_\rho + \mathbf{a}_\phi + \left(\frac{-20}{29} + \frac{10}{\sqrt{29}} \right) \mathbf{a}_z \\ &= 2.467\mathbf{a}_\rho + \mathbf{a}_\phi + 1.167\mathbf{a}_z \end{aligned}$$

Note that at $(-3, 4, 0)$,

$$|\mathbf{B}(x, y, z)| = |\mathbf{B}(\rho, \phi, z)| = |\mathbf{B}(r, \theta, \phi)| = 2.907$$

This may be used to check the correctness of the result whenever possible.

PRACTICE EXERCISE 2.2

Express the following vectors in Cartesian coordinates:

(a) $\mathbf{A} = \rho z \sin \phi \mathbf{a}_\rho + 3\rho \cos \phi \mathbf{a}_\phi + \rho \cos \phi \sin \phi \mathbf{a}_z$

(b) $\mathbf{B} = r^2 \mathbf{a}_r + \sin \theta \mathbf{a}_\phi$

Answer: (a) $\mathbf{A} = \frac{1}{\sqrt{x^2 + y^2}} [(xyz - 3xy) \mathbf{a}_x + (zy^2 + 3x^2) \mathbf{a}_y + xy \mathbf{a}_z]$

(b) $\mathbf{B} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} \{ [x(x^2 + y^2 + z^2) - y] \mathbf{a}_x + [y(x^2 + y^2 + z^2) + x] \mathbf{a}_y + z(x^2 + y^2 + z^2) \mathbf{a}_z \}$

2.5 CONSTANT-COORDINATE SURFACES

Surfaces in Cartesian, cylindrical, or spherical coordinate systems are easily generated by keeping one of the coordinate variables constant and allowing the other two to vary. In the

Cartesian system, if we keep x constant and allow y and z to vary, an infinite plane is generated. Thus we could have infinite planes

$$\begin{aligned}x &= \text{constant} \\y &= \text{constant} \\z &= \text{constant}\end{aligned}\tag{2.34}$$

which are perpendicular to the x -, y -, and z -axes, respectively, as shown in Figure 2.7. The intersection of two planes is a line. For example,

$$x = \text{constant}, \quad y = \text{constant}\tag{2.35}$$

is the line RPQ parallel to the z -axis. The intersection of three planes is a point. For example,

$$x = \text{constant}, \quad y = \text{constant}, \quad z = \text{constant}\tag{2.36}$$

is the point $P(x, y, z)$. Thus we may define point P as the intersection of three orthogonal infinite planes. If P is $(1, -5, 3)$, then P is the intersection of planes $x = 1$, $y = -5$, and $z = 3$.

Orthogonal surfaces in cylindrical coordinates can likewise be generated. The surfaces

$$\begin{aligned}\rho &= \text{constant} \\ \phi &= \text{constant} \\ z &= \text{constant}\end{aligned}\tag{2.37}$$

are illustrated in Figure 2.8, where it is easy to observe that $\rho = \text{constant}$ is a circular cylinder, $\phi = \text{constant}$ is a semiinfinite plane with its edge along the z -axis, and $z = \text{constant}$ is the same infinite plane as in a Cartesian system. Where two surfaces meet is either a line or a circle. Thus,

$$z = \text{constant}, \quad \rho = \text{constant}\tag{2.38}$$

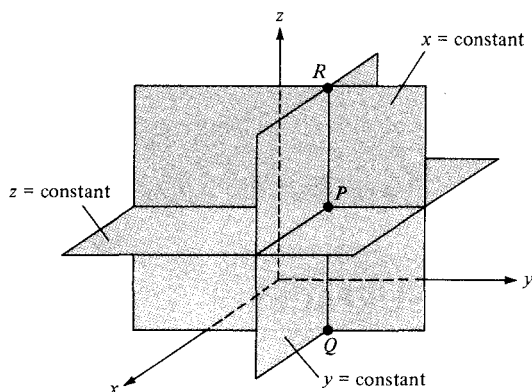
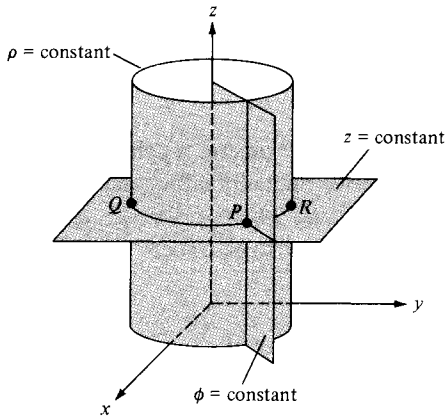


Figure 2.7 Constant x , y , and z surfaces.

Figure 2.8 Constant ρ , ϕ , and z surfaces.

is a circle QPR of radius ρ , whereas $z = \text{constant}$, $\phi = \text{constant}$ is a semiinfinite line. A point is an intersection of the three surfaces in eq. (2.37). Thus,

$$\rho = 2, \quad \phi = 60^\circ, \quad z = 5 \quad (2.39)$$

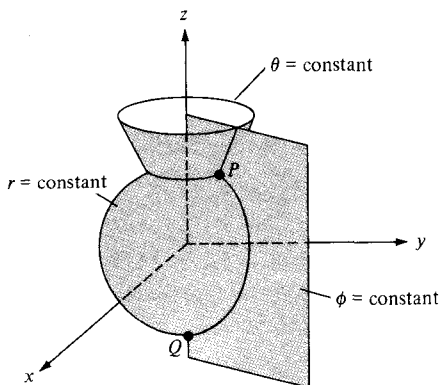
is the point $P(2, 60^\circ, 5)$.

The orthogonal nature of the spherical coordinate system is evident by considering the three surfaces

$$\begin{aligned} r &= \text{constant} \\ \theta &= \text{constant} \\ \phi &= \text{constant} \end{aligned} \quad (2.40)$$

which are shown in Figure 2.9, where we notice that $r = \text{constant}$ is a sphere with its center at the origin; $\theta = \text{constant}$ is a circular cone with the z -axis as its axis and the origin as its vertex; $\phi = \text{constant}$ is the semiinfinite plane as in a cylindrical system. A line is formed by the intersection of two surfaces. For example:

$$r = \text{constant}, \quad \phi = \text{constant} \quad (2.41)$$

Figure 2.9 Constant r , θ , and ϕ surfaces.

is a semicircle passing through Q and P . The intersection of three surfaces gives a point. Thus,

$$r = 5, \quad \theta = 30^\circ, \quad \phi = 60^\circ \quad (2.42)$$

is the point $P(5, 30^\circ, 60^\circ)$. We notice that in general, a point in three-dimensional space can be identified as the intersection of three mutually orthogonal surfaces. Also, a unit normal vector to the surface $n = \text{constant}$ is $\pm \mathbf{a}_n$, where n is x, y, z, ρ, ϕ, r , or θ . For example, to plane $x = 5$, a unit normal vector is $\pm \mathbf{a}_x$ and to plane $\phi = 20^\circ$, a unit normal vector is $\mathbf{a}_\phi$.

EXAMPLE 2.3

Two uniform vector fields are given by $\mathbf{E} = -5\mathbf{a}_\rho + 10\mathbf{a}_\phi + 3\mathbf{a}_z$ and $\mathbf{F} = \mathbf{a}_\rho + 2\mathbf{a}_\phi - 6\mathbf{a}_z$. Calculate

- $|\mathbf{E} \times \mathbf{F}|$
- The vector component of $\mathbf{E}$ at $P(5, \pi/2, 3)$ parallel to the line $x = 2, z = 3$
- The angle $\mathbf{E}$ makes with the surface $z = 3$ at P

Solution:

$$\begin{aligned} \text{(a)} \quad \mathbf{E} \times \mathbf{F} &= \begin{vmatrix} \mathbf{a}_\rho & \mathbf{a}_\phi & \mathbf{a}_z \\ -5 & 10 & 3 \\ 1 & 2 & -6 \end{vmatrix} \\ &= (-60 - 6)\mathbf{a}_\rho + (3 - 30)\mathbf{a}_\phi + (-10 - 10)\mathbf{a}_z \\ &= (-66, -27, -20) \end{aligned}$$

$$|\mathbf{E} \times \mathbf{F}| = \sqrt{66^2 + 27^2 + 20^2} = 74.06$$

- (b) Line $x = 2, z = 3$ is parallel to the y -axis, so the component of $\mathbf{E}$ parallel to the given line is

$$(\mathbf{E} \cdot \mathbf{a}_y)\mathbf{a}_y$$

But at $P(5, \pi/2, 3)$

$$\begin{aligned} \mathbf{a}_y &= \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi \\ &= \sin \pi/2 \mathbf{a}_\rho + \cos \pi/2 \mathbf{a}_\phi = \mathbf{a}_\rho \end{aligned}$$

Therefore,

$$(\mathbf{E} \cdot \mathbf{a}_y)\mathbf{a}_y = (\mathbf{E} \cdot \mathbf{a}_\rho)\mathbf{a}_\rho = -5\mathbf{a}_\rho \quad (\text{or } -5\mathbf{a}_y)$$

- (c) Utilizing the fact that the z -axis is normal to the surface $z = 3$, the angle between the z -axis and $\mathbf{E}$, as shown in Figure 2.10, can be found using the dot product:

$$\mathbf{E} \cdot \mathbf{a}_z = |\mathbf{E}|(1) \cos \theta_{Ez} \rightarrow 3 = \sqrt{134} \cos \theta_{Ez}$$

$$\cos \theta_{Ez} = \frac{3}{\sqrt{134}} = 0.2592 \rightarrow \theta_{Ez} = 74.98^\circ$$

Hence, the angle between $z = 3$ and $\mathbf{E}$ is

$$90^\circ - \theta_{Ez} = 15.02^\circ$$

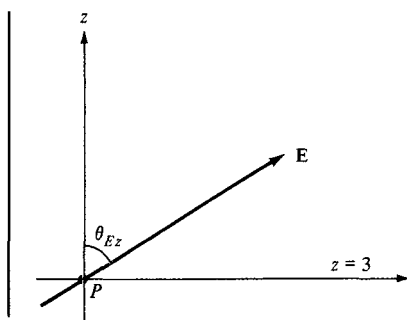


Figure 2.10 For Example 2.3(c).

PRACTICE EXERCISE 2.3

Given the vector field

$$\mathbf{H} = \rho z \cos \phi \mathbf{a}_\rho + e^{-2} \sin \frac{\phi}{2} \mathbf{a}_\phi + \rho^2 \mathbf{a}_z$$

At point $(1, \pi/3, 0)$, find

- (a) $\mathbf{H} \cdot \mathbf{a}_x$
- (b) $\mathbf{H} \times \mathbf{a}_\theta$
- (c) The vector component of $\mathbf{H}$ normal to surface $\rho = 1$
- (d) The scalar component of $\mathbf{H}$ tangential to the plane $z = 0$

Answer: (a) -0.433 , (b) $-0.5 \mathbf{a}_\rho$, (c) $0 \mathbf{a}_\rho$, (d) 0.5 .

EXAMPLE 2.4

Given a vector field

$$\mathbf{D} = r \sin \phi \mathbf{a}_r - \frac{1}{r} \sin \theta \cos \phi \mathbf{a}_\theta + r^2 \mathbf{a}_\phi$$

determine

- (a) $\mathbf{D}$ at $P(10, 150^\circ, 330^\circ)$
- (b) The component of $\mathbf{D}$ tangential to the spherical surface $r = 10$ at P
- (c) A unit vector at P perpendicular to $\mathbf{D}$ and tangential to the cone $\theta = 150^\circ$

Solution:

- (a) At P , $r = 10$, $\theta = 150^\circ$, and $\phi = 330^\circ$. Hence,

$$\mathbf{D} = 10 \sin 330^\circ \mathbf{a}_r - \frac{1}{10} \sin 150^\circ \cos 330^\circ \mathbf{a}_\theta + 100 \mathbf{a}_\phi = (-5, 0.043, 100)$$

(b) Any vector $\mathbf{D}$ can always be resolved into two orthogonal components:

$$\mathbf{D} = \mathbf{D}_t + \mathbf{D}_n$$

where D_t is tangential to a given surface and $\mathbf{D}_n$ is normal to it. In our case, since $\mathbf{a}_r$ is normal to the surface $r = 10$,

$$\mathbf{D}_n = r \sin \phi \mathbf{a}_r = -5\mathbf{a}_r$$

Hence,

$$\mathbf{D}_t = \mathbf{D} - \mathbf{D}_n = 0.043\mathbf{a}_\theta + 100\mathbf{a}_\phi$$

(c) A vector at P perpendicular to $\mathbf{D}$ and tangential to the cone $\theta = 150^\circ$ is the same as the vector perpendicular to both $\mathbf{D}$ and $\mathbf{a}_\theta$. Hence,

$$\begin{aligned} \mathbf{D} \times \mathbf{a}_\theta &= \begin{vmatrix} \mathbf{a}_r & \mathbf{a}_\theta & \mathbf{a}_\phi \\ -5 & 0.043 & 100 \\ 0 & 1 & 0 \end{vmatrix} \\ &= -100\mathbf{a}_r - 5\mathbf{a}_\phi \end{aligned}$$

A unit vector along this is

$$\mathbf{a} = \frac{-100\mathbf{a}_r - 5\mathbf{a}_\phi}{\sqrt{100^2 + 5^2}} = -0.9988\mathbf{a}_r - 0.0499\mathbf{a}_\phi$$

PRACTICE EXERCISE 2.4

If $\mathbf{A} = 3\mathbf{a}_r + 2\mathbf{a}_\theta - 6\mathbf{a}_\phi$ and $\mathbf{B} = 4\mathbf{a}_r + 3\mathbf{a}_\phi$, determine

- $\mathbf{A} \cdot \mathbf{B}$
- $|\mathbf{A} \times \mathbf{B}|$
- The vector component of $\mathbf{A}$ along $\mathbf{a}_z$ at $(1, \pi/3, 5\pi/4)$

Answer: (a) -6 , (b) 34.48 , (c) $-0.116\mathbf{a}_r + 0.201\mathbf{a}_\theta$.

SUMMARY

- The three common coordinate systems we shall use throughout the text are the Cartesian (or rectangular), the circular cylindrical, and the spherical.
- A point P is represented as $P(x, y, z)$, $P(\rho, \phi, z)$, and $P(r, \theta, \phi)$ in the Cartesian, cylindrical, and spherical systems respectively. A vector field $\mathbf{A}$ is represented as (A_x, A_y, A_z) or $A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z$ in the Cartesian system, as (A_ρ, A_ϕ, A_z) or $A_\rho\mathbf{a}_\rho + A_\phi\mathbf{a}_\phi + A_z\mathbf{a}_z$ in the cylindrical system, and as (A_r, A_θ, A_ϕ) or $A_r\mathbf{a}_r + A_\theta\mathbf{a}_\theta + A_\phi\mathbf{a}_\phi$ in the spherical system. It is preferable that mathematical operations (addition, subtraction, product, etc.) be performed in the same coordinate system. Thus, point and vector transformations should be performed whenever necessary.
- Fixing one space variable defines a surface; fixing two defines a line; fixing three defines a point.
- A unit normal vector to surface $n = \text{constant}$ is $\pm \mathbf{a}_n$.

REVIEW QUESTIONS

2.1 The ranges of θ and ϕ as given by eq. (2.17) are not the only possible ones. The following are all alternative ranges of θ and ϕ , except

- (a) $0 \leq \theta < 2\pi, 0 \leq \phi \leq \pi$
- (b) $0 \leq \theta < 2\pi, 0 \leq \phi < 2\pi$
- (c) $-\pi \leq \theta \leq \pi, 0 \leq \phi \leq \pi$
- (d) $-\pi/2 \leq \theta \leq \pi/2, 0 \leq \phi < 2\pi$
- (e) $0 \leq \theta \leq \pi, -\pi \leq \phi < \pi$
- (f) $-\pi \leq \theta < \pi, -\pi \leq \phi < \pi$

2.2 At Cartesian point $(-3, 4, -1)$, which of these is incorrect?

- (a) $\rho = -5$
- (b) $r = \sqrt{26}$
- (c) $\theta = \tan^{-1} \frac{5}{-1}$
- (d) $\phi = \tan^{-1} \frac{4}{-3}$

2.3 Which of these is not valid at point $(0, 4, 0)$?

- (a) $\mathbf{a}_\phi = -\mathbf{a}_x$
- (b) $\mathbf{a}_\theta = -\mathbf{a}_z$
- (c) $\mathbf{a}_r = 4\mathbf{a}_y$
- (d) $\mathbf{a}_\rho = \mathbf{a}_y$

2.4 A unit normal vector to the cone $\theta = 30^\circ$ is:

- (a) $\mathbf{a}_r$
- (b) $\mathbf{a}_\theta$
- (c) $\mathbf{a}_\phi$
- (d) none of the above

2.5 At every point in space, $\mathbf{a}_\phi \cdot \mathbf{a}_\theta = 1$.

- (a) True
- (b) False

2.6 If $\mathbf{H} = 4\mathbf{a}_\rho - 3\mathbf{a}_\phi + 5\mathbf{a}_z$, at $(1, \pi/2, 0)$ the component of $\mathbf{H}$ parallel to surface $\rho = 1$ is

- (a) $4\mathbf{a}_\rho$
- (b) $5\mathbf{a}_z$
- (c) $-3\mathbf{a}_\phi$
- (d) $-3\mathbf{a}_\phi + 5\mathbf{a}_z$
- (e) $5\mathbf{a}_\phi + 3\mathbf{a}_z$

2.7 Given $\mathbf{G} = 20\mathbf{a}_r + 50\mathbf{a}_\theta + 40\mathbf{a}_\phi$, at $(1, \pi/2, \pi/6)$ the component of $\mathbf{G}$ perpendicular to surface $\theta = \pi/2$ is

- (a) $20\mathbf{a}_r$
- (b) $50\mathbf{a}_\theta$
- (c) $40\mathbf{a}_\phi$
- (d) $20\mathbf{a}_r + 40\mathbf{a}_\theta$
- (e) $-40\mathbf{a}_r + 20\mathbf{a}_\phi$

2.8 Where surfaces $\rho = 2$ and $z = 1$ intersect is

- (a) an infinite plane
- (b) a semiinfinite plane
- (c) a circle
- (d) a cylinder
- (e) a cone

2.9 Match the items in the left list with those in the right list. Each answer can be used once, more than once, or not at all.

- | | |
|---|-------------------------|
| (a) $\theta = \pi/4$ | (i) infinite plane |
| (b) $\phi = 2\pi/3$ | (ii) semiinfinite plane |
| (c) $x = -10$ | (iii) circle |
| (d) $r = 1, \theta = \pi/3, \phi = \pi/2$ | (iv) semicircle |
| (e) $\rho = 5$ | (v) straight line |
| (f) $\rho = 3, \phi = 5\pi/3$ | (vi) cone |
| (g) $\rho = 10, z = 1$ | (vii) cylinder |
| (h) $r = 4, \phi = \pi/6$ | (viii) sphere |
| (i) $r = 5, \theta = \pi/3$ | (ix) cube |
| | (x) point |

2.10 A wedge is described by $z = 0, 30^\circ < \phi < 60^\circ$. Which of the following is incorrect:

- (a) The wedge lies in the $x - y$ plane.
- (b) It is infinitely long
- (c) On the wedge, $0 < \rho < \infty$
- (d) A unit normal to the wedge is $\pm \mathbf{a}_z$
- (e) The wedge includes neither the x -axis nor the y -axis

Answers: 2.1b,f, 2.2a, 2.3c, 2.4b, 2.5b, 2.6d, 2.7b, 2.8c, 2.9a-(vi), b-(ii), c-(i), d-(x), e-(vii), f-(v), g-(iii), h-(iv), i-(iii), 2.10b.

PROBLEMS

2.1 Express the following points in Cartesian coordinates:

- (a) $P(1, 60^\circ, 2)$
- (b) $Q(2, 90^\circ, -4)$
- (c) $R(, 45^\circ, 210^\circ)$
- (d) $T(4, \pi/2, \pi/6)$

2.2 Express the following points in cylindrical and spherical coordinates:

- (a) $P(1, -4, -3)$
- (b) $Q(3, 0, 5)$
- (c) $R(-2, 6, 0)$

2.3 (a) If $V = xz - xy + yz$, express V in cylindrical coordinates.
 (b) If $U = x^2 + 2y^2 + 3z^2$, express U in spherical coordinates.

2.4 Transform the following vectors to cylindrical and spherical coordinates:

- (a) $\mathbf{D} = (x + z)\mathbf{a}_y$
- (b) $\mathbf{E} = (y^2 - x^2)\mathbf{a}_x + xyz\mathbf{a}_y + (x^2 - z^2)\mathbf{a}_z$

2.5 Convert the following vectors to cylindrical and spherical systems:

- (a) $\mathbf{F} = \frac{x\mathbf{a}_x + y\mathbf{a}_y + 4\mathbf{a}_z}{\sqrt{x^2 + y^2 + z^2}}$
- (b) $\mathbf{G} = (x^2 + y^2) \left[\frac{x\mathbf{a}_x}{\sqrt{x^2 + y^2 + z^2}} + \frac{y\mathbf{a}_y}{\sqrt{x^2 + y^2 + z^2}} + \frac{z\mathbf{a}_z}{\sqrt{x^2 + y^2 + z^2}} \right]$

2.6 Express the following vectors in Cartesian coordinates:

- (a) $\mathbf{A} = \rho(z^2 + 1)\mathbf{a}_\rho - \rho z \cos \phi \mathbf{a}_\phi$
- (b) $\mathbf{B} = 2r \sin \theta \cos \phi \mathbf{a}_r + r \cos \theta \cos \theta \mathbf{a}_\theta - r \sin \phi \mathbf{a}_\phi$

2.7 Convert the following vectors to Cartesian coordinates:

- (a) $\mathbf{C} = z \sin \phi \mathbf{a}_\rho - \rho \cos \phi \mathbf{a}_\phi + 2\rho z \mathbf{a}_z$
- (b) $\mathbf{D} = \frac{\sin \theta}{r^2} \mathbf{a}_r + \frac{\cos \theta}{r^2} \mathbf{a}_\theta$

2.8 Prove the following:

- (a) $\mathbf{a}_x \cdot \mathbf{a}_\rho = \cos \phi$
 $\mathbf{a}_x \cdot \mathbf{a}_\phi = -\sin \phi$
 $\mathbf{a}_y \cdot \mathbf{a}_\rho = \sin \phi$
 $\mathbf{a}_y \cdot \mathbf{a}_\phi = \cos \phi$
- (b) $\mathbf{a}_x \cdot \mathbf{a}_r = \sin \theta \cos \phi$
 $\mathbf{a}_x \cdot \mathbf{a}_\theta = \cos \theta \cos \phi$
 $\mathbf{a}_y \cdot \mathbf{a}_r = \sin \theta \sin \phi$

$$\begin{aligned}\mathbf{a}_y \cdot \mathbf{a}_\theta &= \cos \theta \sin \phi \\ \mathbf{a}_z \cdot \mathbf{a}_r &= \cos \theta \\ \mathbf{a}_z \cdot \mathbf{a}_\theta &= -\sin \theta\end{aligned}$$

- 2.9** (a) Show that point transformation between cylindrical and spherical coordinates is obtained using

$$r = \sqrt{\rho^2 + z^2}, \quad \theta = \tan^{-1} \frac{\rho}{z}, \quad \phi = \phi$$

or

$$\rho = r \sin \theta, \quad z = r \cos \theta, \quad \phi = \phi$$

- (b) Show that vector transformation between cylindrical and spherical coordinates is obtained using

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta & 0 & \cos \theta \\ \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

or

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

(Hint: Make use of Figures 2.5 and 2.6.)

- 2.10** (a) Express the vector field

$$\mathbf{H} = xy^2z\mathbf{a}_x + x^2yz\mathbf{a}_y + xyz^2\mathbf{a}_z$$

in cylindrical and spherical coordinates.

- (b) In both cylindrical and spherical coordinates, determine $\mathbf{H}$ at $(3, -4, 5)$.

- 2.11** Let $\mathbf{A} = \rho \cos \theta \mathbf{a}_\rho + \rho z^2 \sin \phi \mathbf{a}_z$

- (a) Transform $\mathbf{A}$ into rectangular coordinates and calculate its magnitude at point $(3, -4, 0)$.
 (b) Transform $\mathbf{A}$ into spherical system and calculate its magnitude at point $(3, -4, 0)$.

- 2.12** The transformation $(A_\rho, A_\phi, A_z) \rightarrow (A_x, A_y, A_z)$ in eq. (2.15) is not complete. Complete it by expressing $\cos \phi$ and $\sin \phi$ in terms of x , y , and z . Do the same thing to the transformation $(A_r, A_\theta, A_\phi) \rightarrow (A_x, A_y, A_z)$ in eq. (2.28).

- 2.13** In Practice Exercise 2.2, express $\mathbf{A}$ in spherical and $\mathbf{B}$ in cylindrical coordinates. Evaluate $\mathbf{A}$ at $(10, \pi/2, 3\pi/4)$ and $\mathbf{B}$ at $(2, \pi/6, 1)$.

2.14 Calculate the distance between the following pairs of points:

- (a) $(2, 1, 5)$ and $(6, -1, 2)$
- (b) $(3, \pi/2, -1)$ and $(5, 3\pi/2, 5)$
- (c) $(10, \pi/4, 3\pi/4)$ and $(5, \pi/6, 7\pi/4)$.

2.15 Describe the intersection of the following surfaces:

- (a) $x = 2, \quad y = 5$
- (b) $x = 2, \quad y = -1, \quad z = 10$
- (c) $r = 10, \quad \theta = 30^\circ$
- (d) $\rho = 5, \quad \phi = 40^\circ$
- (e) $\phi = 60^\circ, \quad z = 10$
- (f) $r = 5, \quad \phi = 90^\circ$

2.16 At point $T(2, 3, -4)$, express $\mathbf{a}_z$ in the spherical system and $\mathbf{a}_r$ in the rectangular system.

***2.17** Given vectors $\mathbf{A} = 2\mathbf{a}_x + 4\mathbf{a}_y + 10\mathbf{a}_z$ and $\mathbf{B} = -5\mathbf{a}_\rho + \mathbf{a}_\phi - 3\mathbf{a}_z$, find

- (a) $\mathbf{A} + \mathbf{B}$ at $P(0, 2, -5)$
- (b) The angle between $\mathbf{A}$ and $\mathbf{B}$ at P
- (c) The scalar component of $\mathbf{A}$ along $\mathbf{B}$ at P

2.18 Given that $\mathbf{G} = (x + y^2)\mathbf{a}_x + xz\mathbf{a}_y + (z^2 + zy)\mathbf{a}_z$, find the vector component of $\mathbf{G}$ along $\mathbf{a}_\phi$ at point $P(8, 30^\circ, 60^\circ)$. Your answer should be left in the Cartesian system.

***2.19** If $\mathbf{J} = r \sin \theta \cos \phi \mathbf{a}_r - \cos 2\theta \sin \phi \mathbf{a}_\theta + \tan \frac{\theta}{2} \ln r \mathbf{a}_\phi$ at $T(2, \pi/2, 3\pi/2)$, determine the vector component of $\mathbf{J}$ that is

- (a) Parallel to $\mathbf{a}_z$
- (b) Normal to surface $\phi = 3\pi/2$
- (c) Tangential to the spherical surface $r = 2$
- (d) Parallel to the line $y = -2, z = 0$

2.20 Let $\mathbf{H} = 5\rho \sin \phi \mathbf{a}_\rho - \rho z \cos \phi \mathbf{a}_\phi + 2\rho \mathbf{a}_z$. At point $P(2, 30^\circ, -1)$, find:

- (a) a unit vector along $\mathbf{H}$
- (b) the component of $\mathbf{H}$ parallel to $\mathbf{a}_x$
- (c) the component of $\mathbf{H}$ normal to $\rho = 2$
- (d) the component of $\mathbf{H}$ tangential to $\phi = 30^\circ$

***2.21** Let

$$\mathbf{A} = \rho(z^2 - 1)\mathbf{a}_\rho - \rho z \cos \phi \mathbf{a}_\phi + \rho^2 z^2 \mathbf{a}_z$$

and

$$\mathbf{B} = r^2 \cos \phi \mathbf{a}_r + 2r \sin \theta \mathbf{a}_\phi$$

At $T(-3, 4, 1)$, calculate: (a) $\mathbf{A}$ and $\mathbf{B}$, (b) the vector component in cylindrical coordinates of $\mathbf{A}$ along $\mathbf{B}$ at T , (c) the unit vector in spherical coordinates perpendicular to both $\mathbf{A}$ and $\mathbf{B}$ at T .

***2.22** Another way of defining a point P in space is $(r, \alpha, \beta, \gamma)$ where the variables are portrayed in Figure 2.11. Using this definition, find $(r, \alpha, \beta, \gamma)$ for the following points:

- (a) $(-2, 3, 6)$
- (b) $(4, 30^\circ, -3)$
- (c) $(3, 30^\circ, 60^\circ)$

(Hint: r is the spherical r , $0 \leq \alpha, \beta, \gamma < 2\pi$.)

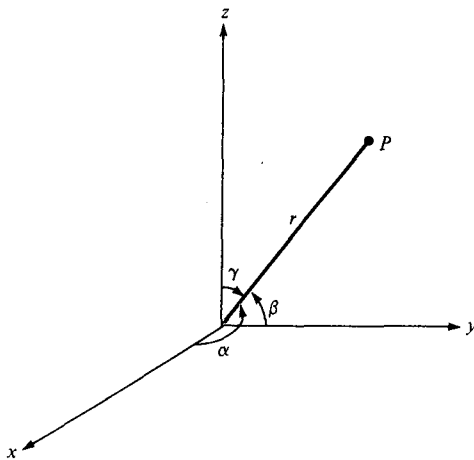


Figure 2.11 For Problem 2.22.

2.23 A vector field in “mixed” coordinate variables is given by

$$\mathbf{G} = \frac{x \cos \phi}{\rho} \mathbf{a}_x + \frac{2yz}{\rho^2} + \left(1 - \frac{x^2}{\rho^2}\right) \mathbf{a}_z$$

Express $\mathbf{G}$ completely in spherical system.